VINAY K R

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LinkedIn | GitHub | Portfolio

SUMMARY

AI Engineer with 3+ years of experience building production software in Python, specializing in Generative AI and Large Language Model (LLM) applications. Experienced in Retrieval-Augmented Generation (RAG), AI agents with LangGraph and Model Context Protocol (MCP), prompt engineering, structured outputs, hybrid search and LLM evaluation. As Founding Engineer at a startup, built a multilingual LLM booking assistant with guardrails and human-in-the-loop approval, and deployed a hybrid-search question-answering service to production; earlier built an LLM-assisted debugging tool at Light & Wonder. Skilled in FastAPI, Pydantic, Docker, PostgreSQL, Redis and OpenTelemetry, with a focus on accuracy, latency and cost.

TECHNICAL SKILLS

- Languages and Frameworks : Python (primary), Go, TypeScript, Node.js, SQL FastAPI, asyncio, Pydantic, REST APIs, Server-Sent Events (SSE), WebSockets, Redis Pub/Sub, React, Angular
- LLM and Agents: LLM APIs (Gemini, Claude, Groq, OpenAI-compatible), prompt engineering, structured outputs (JSON Schema, Pydantic), tool calling, Model Context Protocol (MCP), LangGraph, multi-agent workflows, human-in-the-loop approval, guardrails
- RAG and Search: embeddings, ONNX Runtime, hybrid search, BM25, reciprocal rank fusion, cross-encoder reranking, int8 quantization, chunking, query rewriting, semantic caching
- Document AI: Docling, Optical Character Recognition (OCR), vision-language models (VLMs), schema-constrained extraction, layout-aware parsing
- Evaluation and Observability: held-out benchmarks, Recall@k, MRR, nDCG, F1, LLM-as-a-judge, judge-human agreement, latency and cost tracking, OpenTelemetry, Jaeger, Pytest
- Data and DevOps: PostgreSQL, PostGIS, Redis, SQLite, MySQL, Docker, Kubernetes (k3s), Argo CD, GitHub Actions, GitLab CI/CD

PROJECTS

ReconcileAI: Document AI Reconciliation | Python, FastAPI, Docling, Gemini, Pydantic | GitHub

- Reconciles invoices with purchase orders using Docling parsing, Pydantic-validated Gemini extraction and decimal-exact price, quantity and missing-item checks, each linked to its source bounding box.
- Compared three extraction backends on 20 held-out public CORD receipts: Gemini 3.1 Flash-Lite vision reached 76.0% field F1, vs 66.0% for the Qwen3.8-27B VLM and 56.9% for Docling OCR plus a text LLM.
- Verified the deterministic controls on 50 labelled synthetic invoice/PO pairs across 4 scenarios; routes low-confidence documents to human review and streams batch progress over SSE.

TraceWard: Incident-Triage Agent | Python, FastAPI, LangGraph, MCP, OpenTelemetry | GitHub

- Built a checkpointed LangGraph supervisor that routes crash logs through parsing, similar-incident retrieval, root-cause and fix-drafting specialists, using Model Context Protocol (MCP) tools over stdio.
- Writes a ticket only after an authenticated reviewer approves the exact draft (signed, idempotent, tenant-scoped); enforces step limits and per-run token and cost budgets; traces each step in OpenTelemetry.
- Built a resumable live evaluation (gpt-oss-120b agent, cross-family Qwen judge) on 200 synthetic incidents and 50 held-out cases: root-cause top-k accuracy, tool-call correctness, 20 prompt-injection attacks.

EvidenceRAG: RAG Evaluation Platform | Python, FastAPI, LangGraph, hybrid search | GitHub

- Built a cited-answer RAG pipeline: dense and BM25-style retrieval fused with reciprocal rank fusion, then reranking and context compression, orchestrated with LangGraph.
- Designed SQuAD v1.1 ablations (chunking, dense/BM25/hybrid/reranked retrieval, query rewriting) scored by Recall@k, MRR, nDCG and a cross-family LLM faithfulness judge (Groq Qwen judging GPT-OSS).

PROFESSIONAL EXPERIENCE

Founding Engineer — Mar 2026 – Sep 2026

Medha Innovations | Self-funded, two-founder startup: booking and messaging platform for Pandits and families.

- Built Medha Sahayak, an event-driven LLM booking assistant in Python (FastAPI, Redis Pub/Sub): a guard redacts phone/UPI details, a rule-based router skips small talk, and an LLM extracts Pydantic-validated intent.
- Grounded replies in read-only Panchanga and Pandit-search tools; a grounding gate drops any date, name or price that no tool returned, and suggestions go out only as human-approved cards.
- Built a 60-turn code-mixed English, Kannada and Hindi evaluation set (20 development, 40 held-out) and benchmarked the rule-based baseline against gpt-oss-20b through an OpenAI-compatible adapter.
- Deployed an offline Python documentation QA service: SQLite FTS5 BM25 and ONNX embeddings fused with RRF, cited extractive answers and abstention; held-out recall@5: 90.9% BM25 vs 86.4% fused.
- Cut a cross-encoder reranker's peak memory from 1,420 MiB to 264 MiB (int8 quantization, ONNX Runtime arena off) to fit the 512 MiB container; kept it behind a flag as it added ~1 s against a 250 ms budget.
- Designed the platform's Go modular monolith (21 domains): PostgreSQL row locking against double bookings, PostGIS discovery, RS256 JWT RBAC and Redis Pub/Sub WebSockets, on k3s with Argo CD.

Software Engineer, Full Stack — Nov 2025 – Mar 2026

Liminal Custody | Institutional digital asset custody platform.

- Built a transaction-policy engine for three controls (TRM Labs address-risk screening, FATF Travel Rule checks, transfer velocity limits) as TypeScript APIs with Redis-cached, short-circuit rule evaluation.
- Implemented tenant-level RBAC, quorum approvals, Auth0 authentication and step-up two-factor authentication (2FA) with signature validation, plus TypeORM transactions that prevent partial updates.

Senior Associate Software Engineer — Mar 2023 – Jul 2025

Light & Wonder | Casino management and slot-machine software. Progressed from Intern to Senior Associate.

- Built an internal LLM-assisted debugging tool (Gemini API, React UI), deployed on Microsoft Azure with the DevOps team and used by internal teams, that matched crash reports and telemetry to similar historical hardware incidents and their fixes.
- Selected for the slot-machine modernization team; built an in-game LLM chat assistant (OpenAI API, Angular) that lets guests order food and beverages and choose music from their seat; demoed to management and approved for lab testing.
- Fixed a memory leak that made slot machines hang or restart after about one hour: reproduced it on hardware, corrected component cleanup and resource reuse, and verified sustained operation.
- Built SQL Server-backed APIs and screens for casino cash operations; automated audit capture with triggers and stored procedures; wrote Cypress end-to-end tests.
- Worked on CONNEXUS, the slot-machine tracking and data management system that monitors machine status and usage and collects data on a central server; contributed to performance profiling and monitoring.

EDUCATION

Siddaganga Institute of Technology (SIT), Tumakuru — 2019 – 2023

Bachelor of Engineering (B.E.), Computer Science and Engineering — CGPA: 8.7/10

PUBLICATIONS AND CERTIFICATIONS

"Data Visualisation of Time Tradable Assets Using Machine Learning," IEEE NMITCON — 2023

DOI 10.1109/NMITCON58196.2023.10275962 | ResearchGate

Model Context Protocol (MCP): Introduction and Advanced Topics | Anthropic

Docker Foundations Professional Certificate | LinkedIn Learning